STAFF REVIEW · H446 251 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A stack.

2. O(n).

3. Arithmetic and logical operations.

4. Factor A: A∧(¬B∨B)=A∧1=A. K-map adjacent cells differ only in B, so B is eliminated.

I Do — source worked steps

1. Predict first — Copy PC=50 into MAR to identify the memory address being accessed.

2. Explain the answer — Fetch memory content into MDR, copy it to CIR and advance PC to 51.

We Do — guided CHECK

1. Predict first
MAR must contain 200 and MDR transports the instruction from memory.
Explain the answer
CIR now contains ADD 10; for sequential execution PC has advanced to 201.

You Do — source mark schemes

1. [5] PC next instruction address [1], MAR accessed address [1], MDR transferred data/instruction [1], CIR current instruction [1], ACC intermediate arithmetic/logical result [1].

2. [6] PC initially 73 [1], MAR becomes 73 [1], memory transfers LDA 92 via MDR [1], CIR receives LDA 92 [1], sequential PC becomes 74 [1], CU interprets the instruction [1].

3. [6] MAR carries memory location address [1]; MDR carries actual contents [1]; fetching uses PC→MAR [1] and memory→MDR→CIR [1]; loading data uses an address in MAR [1] and the retrieved value via MDR towards target register/ACC [1].

Exit Ticket — source answer

1. False. MAR holds the address; MDR carries the contents.